Pure Y1 Ch 5 Straight Line G

Starter (Pre-requisites)

1) Find the point of intersections of the following pairs of lines.
   a) $y = 4x + 7$ and $3y = 2x - 1$ $(-2, -1)$
   b) $y = 5x - 1$ and $3x + 7y = 11$ $\left(\frac{9}{19}, \frac{26}{19}\right)$
   c) $2x - 5y = -1$ and $5x - 7y = 14$ $(7, 3)$

2) Simplify each of the following:
   a) $\sqrt{80}$ $4\sqrt{5}$
   b) $\sqrt{200}$ $10\sqrt{2}$
   c) $\sqrt{125}$ $5\sqrt{5}$

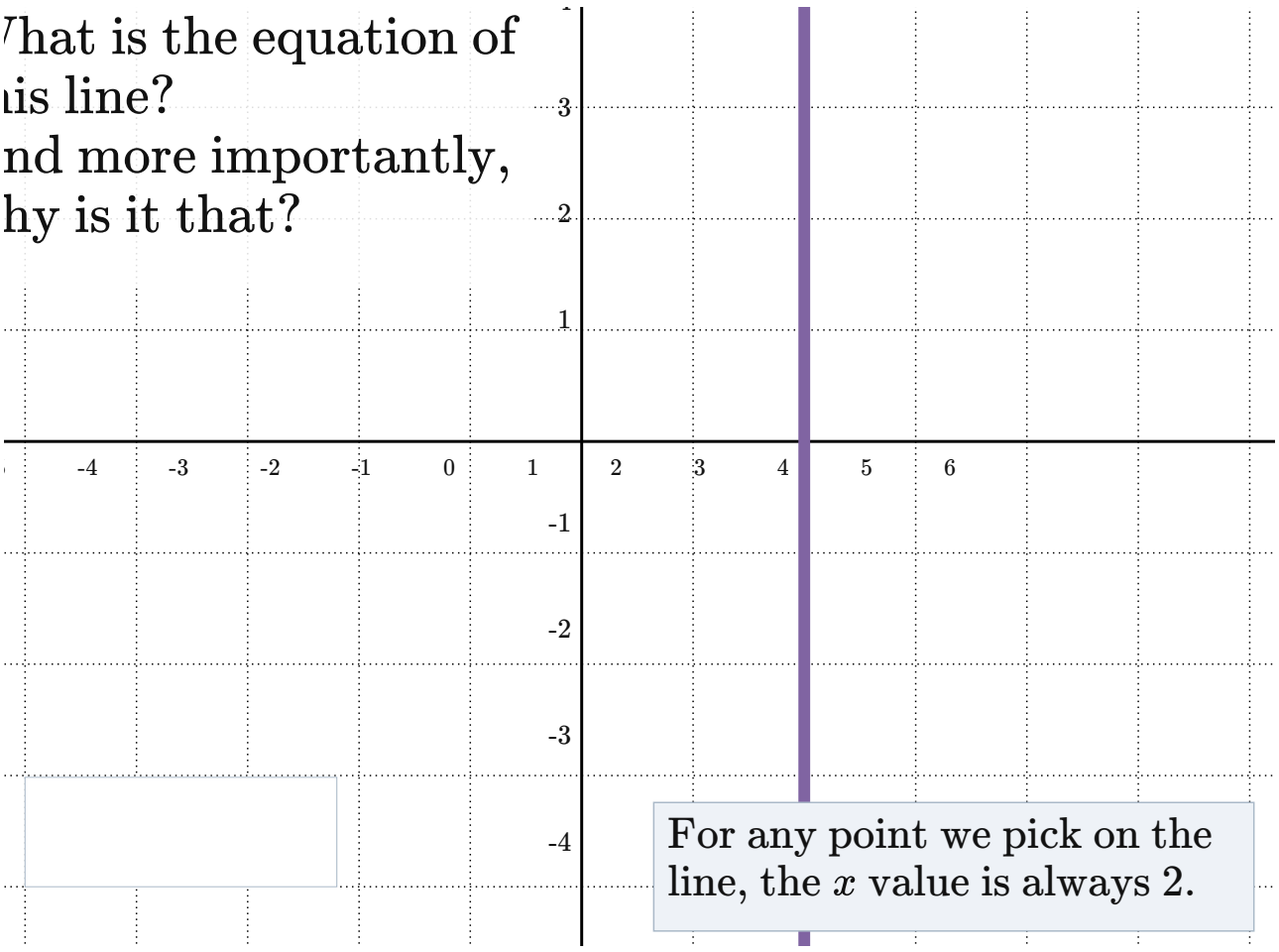
3) Make y the subject of each equation:
   a) $6x + 3y - 15 = 0$ $y = 5 - 2x$
   b) $2x - 5y - 9 = 0$ $y = \frac{2}{5}x - \frac{9}{5}$
   c) $3x - 7y + 12 = 0$ $y = \frac{3}{7}x + \frac{12}{7}$

1: $y = mx + c$, Gradient & Determining Equations

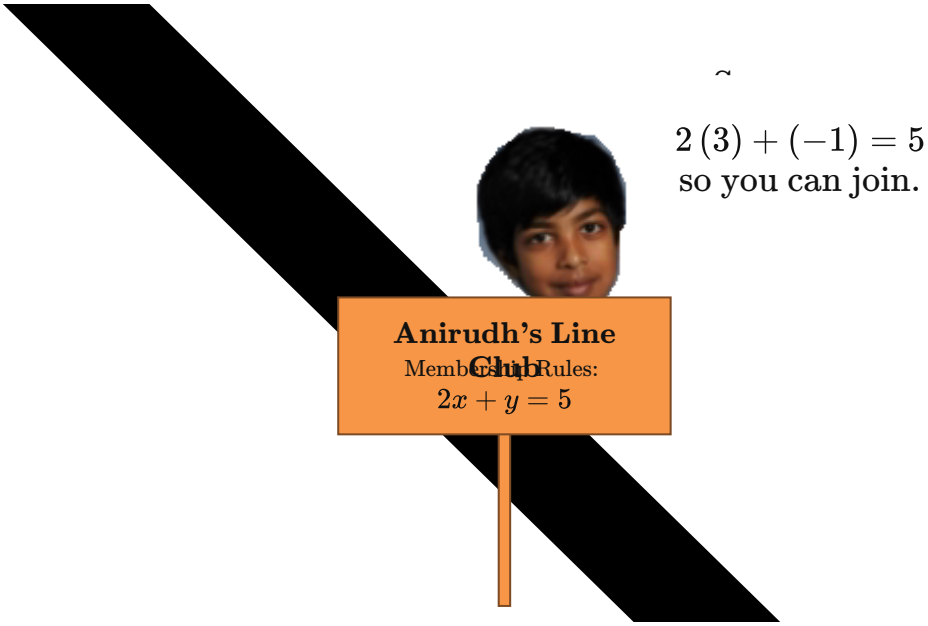
2: Parallel/Perpendicular Lines

3: Lengths and Areas

4: Modelling



! A line consists of all points which satisfy some equation in terms of x and/or y.



This means we can **substitute** the values of a coordinate into our equation whenever we know the point lies on the line.

The point $(5, a)$ lies on the line with equation $y = 3x + 2$. Determine the value of a.

Find the coordinate of the point where the line $2x + y = 5$ cuts the x-axis.

Determine where the line $x + 2y = 3$ crosses the:

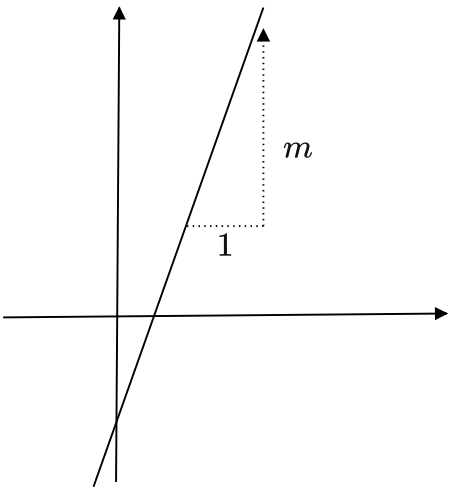
a) y-axis:

b) x-axis:

What mistakes do you think it's easy to make?

-
-

The steepness of a line is known as the **gradient**.
It tells us what y changes by as x increases by 1.



So if the y value increased by 6 as the x value increased by 2, what is y increasing by for each unit increase of x?
How would that give us a suitable formula for the gradient m?

$$m =$$

Δ is the (capital) Greek letter "delta" and means **"change in"**.

Textbook Note:

The textbook uses $m = \frac{y_2 - y_1}{x_2 - x_1}$ for two points (x_1, y_1) and (x_2, y_2). Reasons I don't use it for non-algebraic coordinates:

- Students often get the y_1 and y_2 the wrong way round (or with the x's)
- Students often make sign errors when dealing with negatives, e.g. $(-3) - (-4)$
- It can't be done as easily mentally,
- Students see it as "yet another formula to learn" when really all you need is to appreciate is what gradient is, i.e. "y change per x change".

Find the gradient of the line that goes through the points:

1 $(1, 4)$ $(3, 10)$ $m =$

2 $(5, 7)$ $(8, 1)$ $m =$

3 $(2, 2)$ $(-1, 10)$ $m =$

4 Show that the points A $(3, 4)$, $B\,(5, 5)$, $C\,(11, 8)$ all lie on a straight line.

$m_{AB} = \frac{1}{2}$ $m_{BC} = \frac{3}{6} = \frac{1}{2}$

The line joining $(2, -5)$ to $(4, a)$ has gradient -1. Work out the value of a.

One form we can put a straight line equation in is:

$$y = mx + c$$

Gradient

y-intercept

Determine the gradient and y-intercept of the line with equation $4x - 3y + 5 = 0$

Make y the subject so we have the form
$y = mx + c$
Put y on the side it's positive.

Divide **each** term by 3; don't write $y = \frac{4x+5}{3}$ otherwise it's not in the form $y = mx + c$

This is algebra, so use improper fractions, and not mixed numbers or recurring decimals.

At GCSE, **y=mx+c** was the main form you would express a straight line equation, sometimes known as the '**slope-intercept form**'.
But another common form is **ax+by+c=0**, where a, b, c are integers. This is known as the '**standard**' form.

Express $y = \frac{1}{3}x - \frac{2}{3}$ in the form $ax + by + c = 0$, where a, b, c are integers.

We'll see on the next slide WHY we might want to put an equation in this form over $y = mx + c$...

Why might we want to put a straight line equation in the form $ax + by + c = 0$?

$y = mx + c$
"Slope-Intercept Form"

$ax + by + c = 0$
"Standard Form"

Coverage

$y = mx + c$ doesn't allow you to represent vertical lines. Standard form allows us to do this by just making b zero.

$x + 4 = 0$

Symmetry

In general, the '**linear combination**' of two variables x and y is $ax + by$, i.e. "some amount of x and some amount of y". There is a greater elegance and symmetry to this form over $y = mx + c$ because x and y appear similarly within the expression.

Usefulness

This more 'elegant' form also means it ties in with vectors and matrices. In FM, you will learn about the '**dot product**' of two vectors:

$$\begin{pmatrix} a \\ b \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} = ax + by$$

thus since $ax + by + c = 0$, we can represent a straight line using:

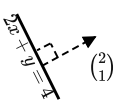
$$\begin{pmatrix} a \\ b \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} + c = 0 \qquad (1)$$

We can extend to 3D points to get the equation of a **plane**:

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} + d = 0 \qquad (2)$$

Conveniently, in equation (1), the vector $\begin{pmatrix} a \\ b \end{pmatrix}$ is **perpendicular to the line**. And in equation (2), the vector $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$ is perpendicular to the plane.
Nice!

$2x + y = 4$ ➡ $\begin{pmatrix} 2 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} = 4$ ➡



Express $y = \frac{2}{5}x + \frac{3}{5}$ in the form $ax + by + c = 0$, where a, b, c are integers.

9 Show that the points $A(2, 3)$, $B(4, 4)$ and $C(10, 7)$ can be joined by a straight line.

10 Show that the points $A(-2a, 5a)$, $B(0, 4a)$ and points $C(6a, a)$ are collinear. (3 ma...

8 A line l cuts the x-axis at $(5, 0)$ and the y-axis at $(0, 2)$.

a Find the gradient of the line.

b Find an equation of the line in the form $ax + by + c = 0$.

9 Show that the line with equation $ax + by + c = 0$ has gradient $-\frac{a}{b}$ and cuts the y-axis at $-\frac{c}{b}$.

10 The line l with gradient 3 and y-intercept $(0, 5)$ has the equation $ax - 2y + c = 0$.

Find the values of a and c. (2 marks)

11 The straight line l passes through $(0, 6)$ and has gradient -2. It intersects the line with equation $5x - 8y - 15 = 0$ at point P. Find the coordinates of P.

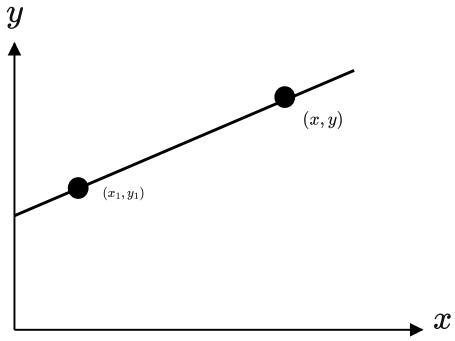
12 The straight line l_1 with equation $y = 3x - 7$ intersects the straight line l_2 with equation $ax + 4y - 17 = 0$ at the point $P(-3, b)$.

a Find the value of b.

b Find the value of a.

Find the equation of the line that goes through $(3, 5)$ and has gradient 2.

How would you have done this at **GCSE**?



Notes: Note that x_1 and y_1 are constants while x and y are variables. The latter are variables because as these 'vary', we get different points on the line.

Suppose that (x_1, y_1) is some fixed point on the line that we specify (e.g. $(3, 5)$). Suppose that (x, y) represents a generic point on the line, which is allowed to change as we consider different points on this line.
Then:

$m = \frac{y-y_1}{x-x_1}$
Thus:

✎ The equation of a line that has gradient m and passes through a point (x_1, y_1) is:

$$y - y_1 = m(x - x_1)$$

Let's revisit:

Find the equation of the line that goes through $(3, 5)$ and has gradient 2.

In a nutshell: You can use this formula whenever you have (a) a gradient and (b) any point on the line.

Gradient	Point	(Unsimplified)
3	$(1, 2)$	
5	$(3, 0)$	
2	$(-3, 4)$	
$\frac{1}{2}$	$(1, -5)$	
9	$(-4, -4)$	

Important Side Note: I've found that many students shun this formula and just use the GCSE method. Please persist with it – it'll be much easier when fractions are involved. Further Mathematicians, don't even think about using the GCSE method, because you'll encounter massive headaches when you consider algebraic points. Trust me on this one!

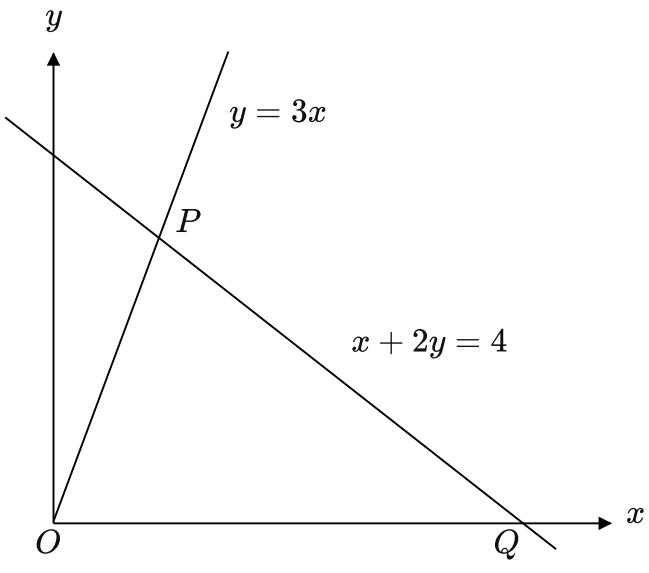
Find the equation of the line that goes through $(4, 5)$ and $(6, 2)$, giving your equation in the form $ax + by + c = 0$.

Test Your Understanding:
Find the equation of the line that goes through $(-1, 9)$ and $(4, 5)$, giving your equation in the form $ax + by + c = 0$.

3 Find the equation of the line l which passes through the points $A(7, 2)$ and $B(9, -8)$. Give your answer in the form $ax + by + c = 0$.

5 T l $(a, 4)$ $(3a, 3)$. An equation of l is $x + 6y + c = 0$. Find the value of a and the value of c.

6 The straight line l passes through $(7a, 5)$ and $(3a, 3)$. An equation of l is $x + by - 12 = 0$. Find the value of a and the value of b.



The diagram shows two lines with equations $y = 3x$ and $x + 2y = 4$, which intersect at the point P.

a) Determine the coordinates of P.

When y=0,x=4
Area $= \frac{1}{2} \times 4 \times \frac{12}{7} = \frac{24}{7}$

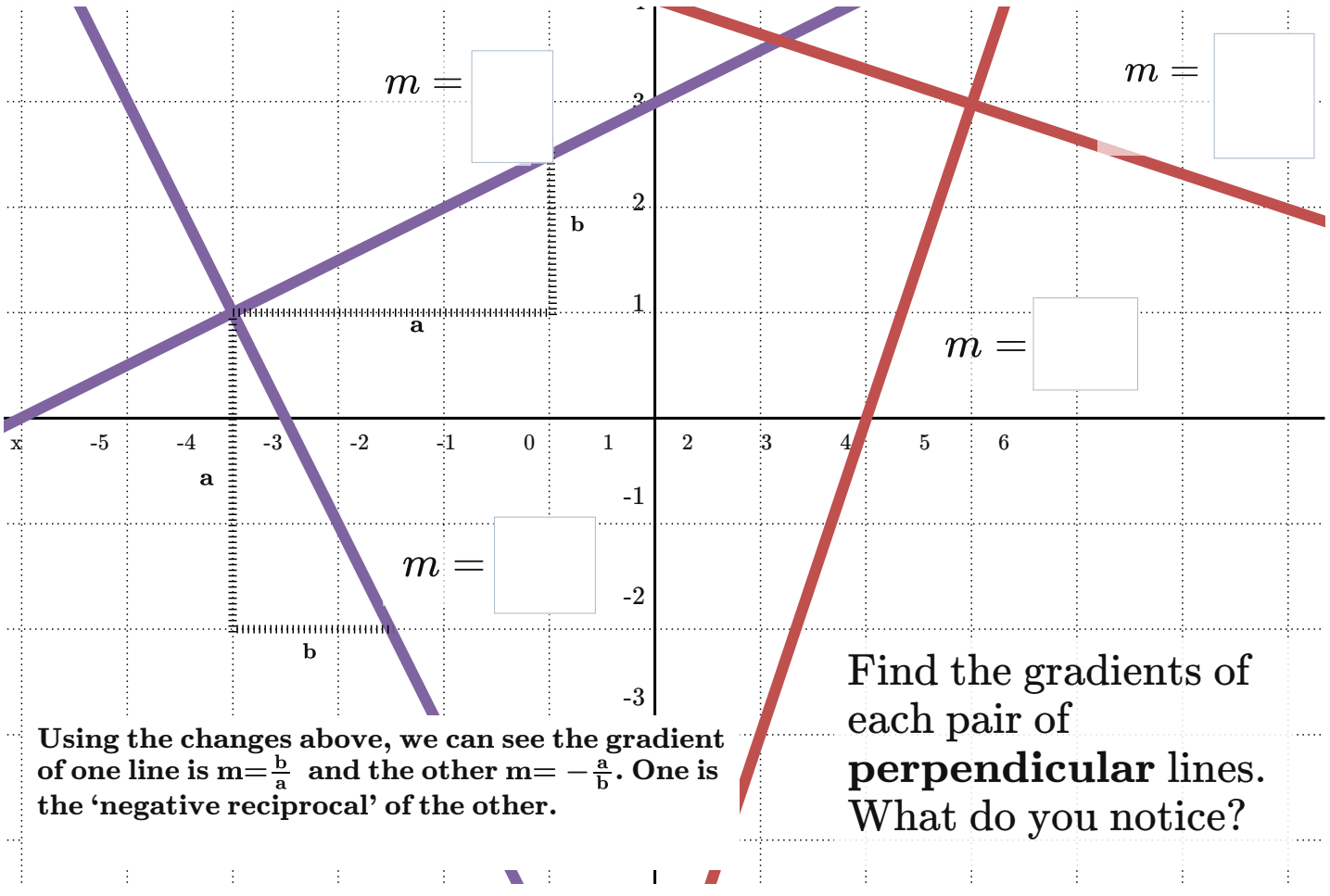
The straight line L_1 passes through the points $(-1, 3)$ and $(11, 12)$.
(a) Find an equation for L_1 in the form $ax + by + c = 0$, where a, b and c are integers. (4)
The line L_2 has equation $3y + 4x - 30 = 0$.
(b) Find the coordinates of the point of intersection of L_1 and L_2. (3)

6 The line that passes through the points $(-3, -5)$ and $(4, 9)$ meets the y-axis at the point G. Work out the coordinates of the point G.

9 The lines $y = 4x - 10$ and $y = x - 1$ intersect at the point T. Find the equation of the line with gradient $-\frac{2}{3}$ that passes through the point T. Write your answer in the form $ax + by + c = 0$, where a, b and c are integers.

14 The line $y = 4x + 5$ meets the y-axis at the point C. The line $y = -3x - 15$ meets the x-axis at the point D. Find the equation of the line joining the points C and D. Write your answer in the form $ax + by + c = 0$, where a, b and c are integers.

16 The lines $y = -2x + 1$ and $y = x + 7$ intersect at the point L. The point M has coordinates $(-3, 1)$. Find the equation of the line that passes through the points L and M.



Using the changes above, we can see the gradient of one line is m=$\frac{b}{a}$ and the other m= $-\frac{a}{b}$. One is the 'negative reciprocal' of the other.

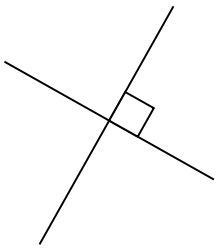
Find the gradients of each pair of **perpendicular** lines. What do you notice?

The gradients of parallel lines are equal.
If two lines are perpendicular, then the gradient of one is the **negative reciprocal** of the other.

$$m_1 = -\frac{1}{m_2}$$

To **show** that two lines are perpendicular:

$$m_1 m_2 = -1$$



Gradient	
2	
-3	
$\frac{1}{4}$	
5	
$-\frac{2}{7}$	
$\frac{7}{5}$	

1 A line is goes through the point (9,10) and is perpendicular to another line with equation $y = 3x + 2$. What is the equation of the line?

2 A line L_1 goes through the [illegible] second line L_2 is perpendicular to L_1 and passes through point B. Where does L_2 cross the x-axis?

(5,0)

3 Are the following lines para[illegible]

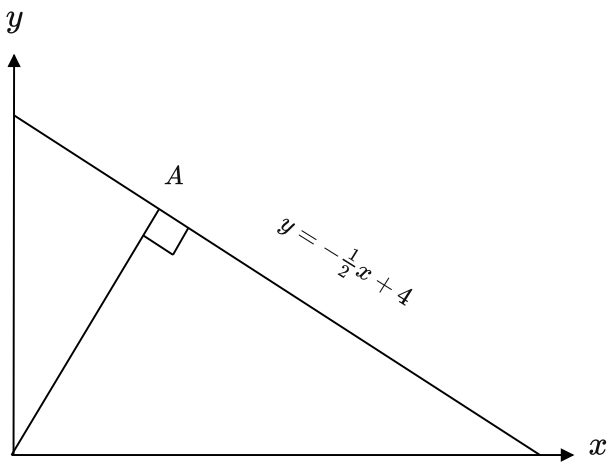
$$2x - y + 4 = 0$$

Neither. Gradients are $\frac{1}{2}$ and 2. But $\frac{1}{2} \times 2 = 1$, not -1, so not perpendicular.

1 A line goes through the point (4,7) and is perpendicular to another line with equation $y = 2x + 2$. What is the equation of the line? Put your answer in the form $ax + by + c = 0$, where a, b, c are integers.

$\mathbf{y = -\frac{1}{2}x + 9}$

2 Determine the point A.



3 The coordinates of a quadrilateral $ABCD$ are $A(-6, 2), B(4, 8), C(6, 1)$ and $D(-9, -8)$. Show that the quadrilateral is a trapezium.

7 A line is parallel to the line $2x - 3y - 1 = 0$ and it passes through the point $(0, 0)$. Write down the equation of the line.

9 The line l passes through the points $(-3, 0)$ and $(3, -2)$ and the line n passes through the points $(1, 8)$ and $(-1, 2)$. Show that the lines l and n are perpendicular.

10 The vertices of a quadrilateral $ABCD$ have coordinates $A(-1, 5), B(7, 1), C(5, -3)$ and $D(-3, 1)$. Show that the quadrilateral is a rectangle.

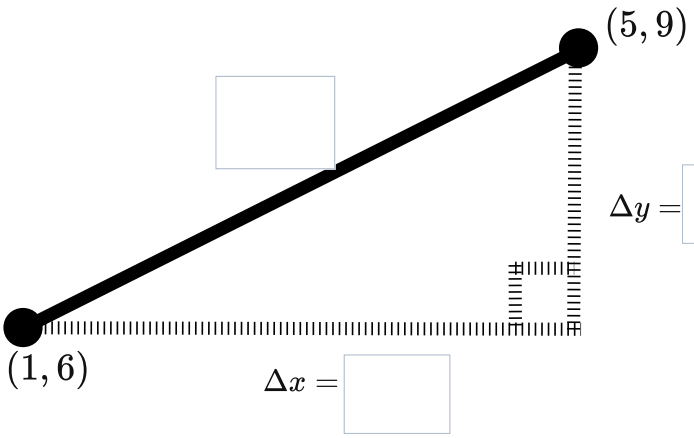
Extension Problems

1

2 *[MAT 2014 1D]* The reflection of the point $(1, 0)$ in the line $y = mx$ has coordinates: (in terms of m)

3 *[STEP I 2004 Q6]* The three points A, B, C have coordinates $(p_1, q_1), (p_2, q_2)$ and (p_3, q_3), respectively. Find the point of intersection of the line joining A to the midpoint of BC, and the line joining B to the midpoint of AC. Verify that this point lies on the line joining C to the midpoint of AB.
The point H has coordinates $(p_1 + p_2 + p_3, q_1 + q_2 + q_3)$. Show that if the line AH intersects the line BC at right angles, then $p_2^2 + q_2^2 = p_3^2 + q_3^2$, and write down a similar result if the line BH intersects the line AC at right angles.
Deduce that if AH is perpendicular to BC and also BH is perpendicular to AC, then CH is perpendicular to AB.

Recall: Δ (said 'delta') means "change in".



How could we find the **distance** between these two points?

Hint

Distance between two points:

$$\sqrt{(\Delta x)^2 + (\Delta y)^2}$$

Distance between:

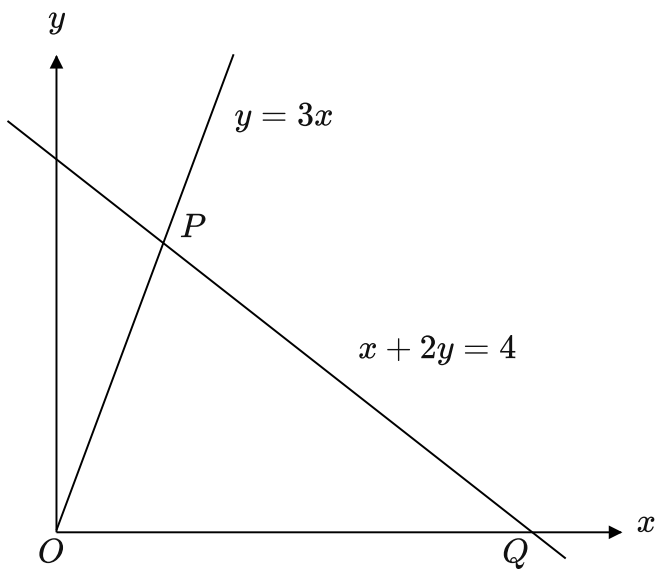
$(3, 4)$ and $(5, 7)$
$(5, 1)$ and $(6, -3)$
$(0, -2)$ and $(-1, 3)$

Note: Unlike with gradient, we don't care if the difference is positive or negative (it's being squared to make it positive anyway!)

Quickfire Questions:

Distance between:

$(1, 10)$ and $(4, 14)$
$(3, -1)$ and $(0, 1)$
$(-4, -2)$ and $(-12, 4)$



The diagram shows two lines with equations $y = 3x$ and $x + 2y = 4$, which intersect at the point P.

a) Determine the coordinates of P.

(We did this in a previous lesson)

Just solve two equations simultaneously.

$\mathbf{x+2\,(3x)=4}$

$\mathbf{7x=4}$

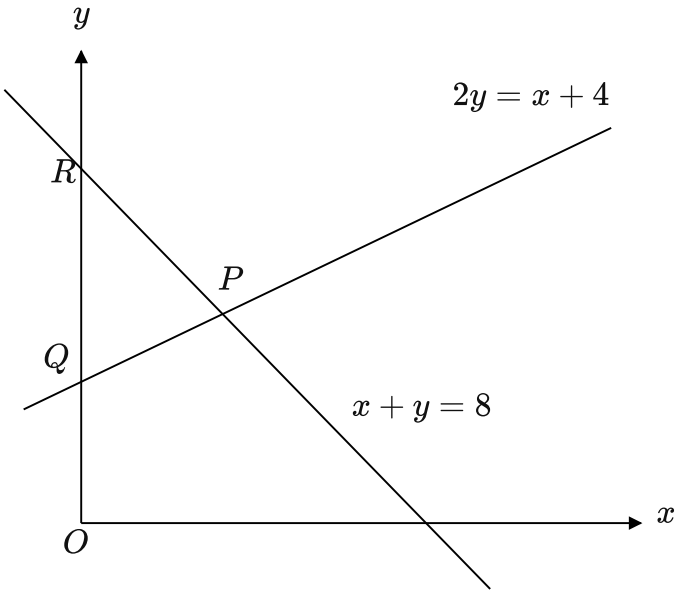
$\mathbf{x=\frac{4}{7}}$

$\mathbf{y=3\,(\frac{4}{7})=\frac{12}{7}}$

b) The line $x + 2y = 4$ intersects the x-axis at the point Q. Determine the area of the triangle OPQ.

When y=0,x=4

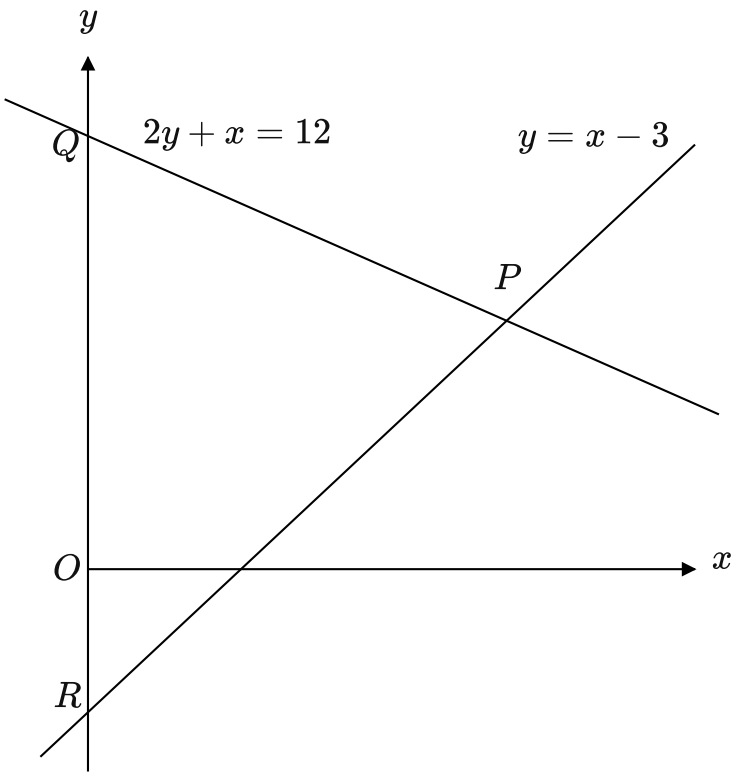
Area $\mathbf{=\frac{1}{2}\times 4\times \frac{12}{7}=\frac{24}{7}}$



Tip: When finding areas of triangles in exam questions, one line is often vertical or horizontal. You should generally choose this to be the 'base' of your triangle.

a

b



a) Determine the coordinate of P.

2 (

x=

b) Determine the area of PQR.

R (0, −3)

Q

Ar

c)

P (

Dist $= \sqrt{6^2+3^2} = \sqrt{45}$

7 A point P lies on the line with equation $y = 4 - 3x$. The point P is a distance $\sqrt{34}$ from the origin. Find the two possible positions of point P. (5 ma...

8 The vertices of a triangle are $A(2, 7), B(5, -6)$ and $C(8, -6)$.

a Show that the triangle is a scalene triangle.

b Find the area of the triangle ABC.

12 The straight line l_1 passes through the point $(-4, 14)$ and has gradient $-\frac{1}{4}$.
a Find an equation for l_1 in the form $ax + by + c = 0$, where a, b and c are integers.
b Write down the coordinates of A, the point where straight line l_1 crosses the y-axis.
The straight line l_2 passes through the origin and has gradient 3. The lines l_1 and l_2 intersect at the point B.
c Calculate the coordinates of B.
d Calculate the exact area of $\triangle OAB$.

Extension Problems

1

We saw in Chapter 2 that lots of things in real life have a 'quadratic' relationship, e.g. vertical height with time. Lots of real life variables have a 'linear' relationship, i.e. **there is a fixed increase/decrease in one variable each time the other variable goes up by 1 unit.**

Examples

Car sales made and take home pay.



The relationship between Celsius and Fahrenheit.

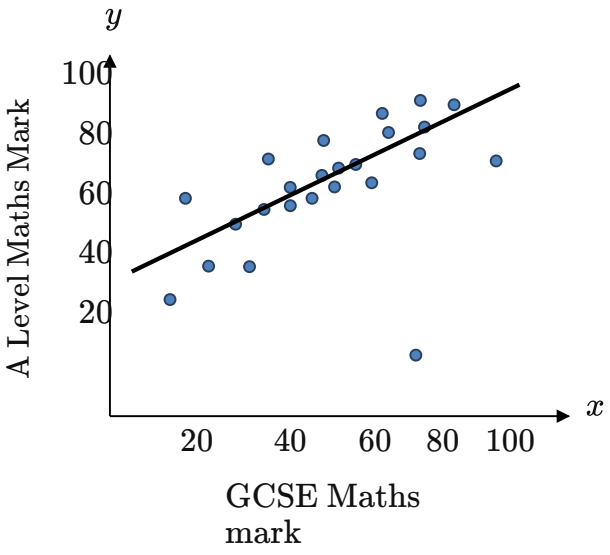


Temperature and altitude (in a particular location)



(And a pure maths one:) The nth term of an arithmetic series.

3, 5, 8, 11, 14, . . .



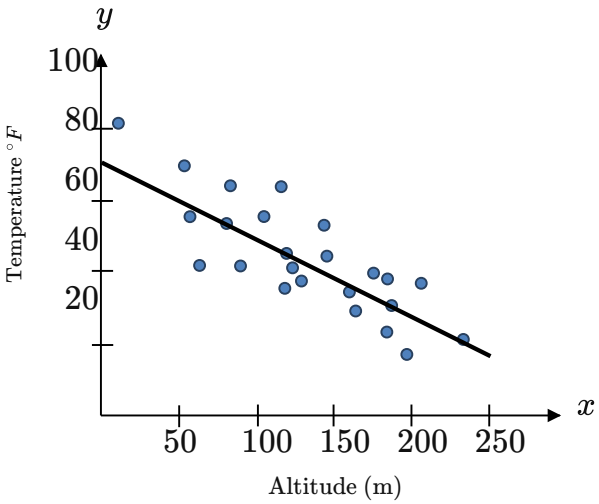
A mathematical model is an attempt to model a real-life situation based on mathematical concepts.

For this example, **our model might be a linear model with equation y=mx+c** where x is a student's GCSE mark and y is the predicted A Level mark.

Such a linear model can be drawn as a line of best fit.
The data obviously doesn't fit this line exactly. This chosen model may only partially fit the data (and the further the points are away from the line, the less suitable this model is).
We might decide another model, e.g. $y = ax^2 + bx + c$, is more appropriate.

But if we choose a well-known model such as a linear one, then we can use **established mathematical theory** in useful ways:

- We need to choose the most appropriate 'parameters' m and c so the model best matches the data. You will learn in S1 there are existing techniques to do this.
- We can then predict a student's A Level mark based on their maths mark.



The temperature y at different points on a mountain is recorded at different altitudes x. Suppose we were to use a linear model $y = mx + c$.

---

a Determine m and c (you can assume the line goes through $(0, 70)$ and $(250, 20)$).

Interpret the meaning o

The temperature at se **s down 0.2°F for each**

b

c

Choosing a model for our data usually involves making **simplifying assumptions**. For a linear model we are assuming:

The y value goes up the same amount for each unit increase in x.

The current population of Bickerstonia is 26000. This year (2017) the population increased by 150. Matt decides to model the population P based on the years t after 2017 by the linear model:

$$P = mt + c$$

Why might this not be a suitable model?

3 The cost of electricity, E, in pounds and the number of kilowatt hours, h, are shown in the table.

kilowatt hours, h	0	15	40	60	80	110
cost of electricity, E	45	46.8	49.8	52.2	54.6	58.2

a Draw a graph of the data.
b Explain how you know a linear model would be appropriate.
c Deduce an equation in the form $E = ah + b$.
d Interpret the meaning of the coefficients a and b.
e Use the model to find the cost of 65 kilowatt hours.

6 The average August temperature in Exeter is 20°C or 68°F. The average January temperature in the same place is 9°C or 48.2°F.
a Write an equation linking Fahrenheit F and Celsius C in the form $F = aC + b$. (3 marks)
b Interpret the values of a and b. (2 marks)
c The highest temperature recorded in the UK was 101.3°F. Calculate this temperature in Celsius.

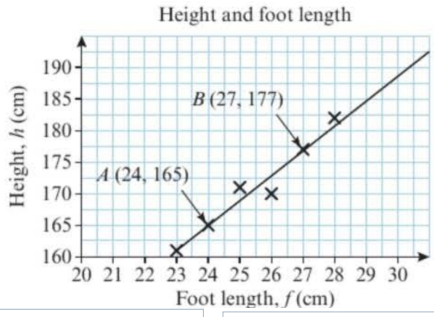
d For what value is the temperature in Fahrenheit the same as the temperature in

8 The scatter graph shows the height h and foot length f of 8 students. A line of best fit is drawn on the scatter graph.

a Explain why the data can be approximated to a linear model. (1 mark)

b Use points A and B on the scatter graph to write a linear equation in the form $h = af + b$. (3 marks)

c Calculate the expected height of a person with a foot length of 26.5 cm. (1 mark)



9 The price P of a good and the quantity Q of a good are linked.

The demand for a new pair of trainers can be modelled using the equation $P = -\frac{3}{4}Q + 35$.

The supply of the trainers can be modelled using the equation $P = \frac{2}{3}Q + 1$.

a Draw a sketch showing the demand and supply lines on the same pair of axes.

The equilibrium point is the point where the supply and demand lines meet.

b Find the values of P and Q at the equilibrium point.